Type MS · Read an enzyme optimum from data and judge how reliable it is

Answer 1 of 11 · Question 4 part a.iii · 1 mark

2024 JUN · Paper 01 (regional R) · find part a.iii in this mark scheme extract

4 (a)	(i)	• axes labels and units and right way around; • linear scale and at least half grid; • correctly plotted points to half square tolerance;		(3)
	(ii)	• curve of best fit;	do not accept straight line going through all points	(1)
	(iii)	6		(1)
	(iv)	• time taken to digest starch decreases until pH 6; • time taken to digest starch increases (from/after PH6) to pH10;	Allow 1 mark for decrease and then increase in time as pH increases	(2)
	(v)	• reference to pH2 not being optimum/best pH for **enzyme** (activity)/pH6 being (closer to) optimum pH for enzyme; • enzymes denatured / active site changes shape/deformed (at pH2); • less/no binding to substrate/ substrate cannot fit into active site/ fewer enzyme -substrate complex;		(3)
	(vi)	small intestine/ mouth/duodenum;		(1)
(b)		• add iodine solution; • blue-black/blue/black colour indicates the presence of starch;		(2)

Total for Question 4 = 13 marks

Type MS · Read an enzyme optimum from data and judge how reliable it is

Answer 2 of 11 · Question 7 part b.i · 1 mark

2019 JUN · Paper 01 (regional R) · extract 1 of 3 · find part b.i in this mark scheme extract

7(a)	Any four from: • volume of lactose/sugar solution used; • concentration of lactose/sugar solution; • volume of enzyme solution used; • concentration of enzyme; • type of enzyme used; • how to measure amount/concentration of lactic acid / apparatus to measure amount / concentration of lactic acid;	Allow amount	**(4)**

Question Number	Answer	Additional guidance	Mark
7(b)(i)	25 °C;	accept 20 °C to 45 °C	**(1)**

Question Number	Answer	Additional guidance	Mark
7(b)(ii)	Any two from: • too few results/temperatures investigated / temperatures are too far apart/large gaps between temperatures; • peak/maximum lactic acid production could be below 25°C / could be above 25°C;		**(2)**

Question Number	Answer	Additional guidance	Mark
7(c)	Any three from: • little/no increase in lactic acid concentration / lactic acid concentration would reach a maximum/peak/become constant; • enzyme denatures (due to decrease in pH); • all substrate converted/broken down/used up; • so reaction /slows down/stops;		**(3)**

Type MS · Read an enzyme optimum from data and judge how reliable it is

Answer 2 of 11 · Question 7 part b.i · 1 mark

2019 JUN · Paper 01 (regional R) · extract 2 of 3 · find part b.i in this mark scheme extract

Question Number	Answer	Additional guidance	Mark
7(d)	Any five from: • not possible (to be completely certain); • due to (low) number of pH/temperature values tested/more pH/temperature values need to be tested; • pH5 appears to be optimum pH; • because most sugar broken down/converted at this pH; • optimum pH could be between 5-8/2-5; • optimum temperature not known; • no repeats carried out/no means calculated; • anomalies not identified; • there could be other components in milk that increase enzyme activity/control variables not taken into account;	Allow value stated within ranges given	**(5)**

Total 15 Marks

Type MS · Read an enzyme optimum from data and judge how reliable it is

Answer 2 of 11 · Question 7 part b.i · 1 mark

2019 JUN · Paper 01 (regional R) · extract 3 of 3 · find part b.i in this mark scheme extract

Paper 1: Subject Human Biology

Question Number	Answer	Additional guidance	Mark

Type MS · Read an enzyme optimum from data and judge how reliable it is

Answer 3 of 11 · Question 4 part a · 2 marks

2020 NOV · Paper 02 (regional R) · extract 1 of 2 · find part a in this mark scheme extract

Question number				
4	(a)	• bell-shaped line (1) • peak at 37°C ± 3°C (1)		2
	(b)	• increase in temperature (1) • molecules have more (kinetic) energy (1) • more collisions (1) • more enzyme-substrate complexes formed (1) • greater rate of activity (1)		5
	(c)	• hot (1) • because optimum temperature is between 60-75°C (1)		2

Total for this question = 9 marks

Type MS · Read an enzyme optimum from data and judge how reliable it is

Answer 3 of 11 · Question 4 part a · 2 marks

2020 NOV · Paper 02 (regional R) · extract 2 of 2 · find part a in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Read an enzyme optimum from data and judge how reliable it is

Answer 4 of 11 · Question 4 part c · 2 marks

2020 NOV · Paper 02 (regional R) · extract 1 of 2 · find part c in this mark scheme extract

Question number				
4	(a)	• bell-shaped line (1) • peak at 37°C ± 3°C (1)		2
	(b)	• increase in temperature (1) • molecules have more (kinetic) energy (1) • more collisions (1) • more enzyme-substrate complexes formed (1) • greater rate of activity (1)		5
	(c)	• hot (1) • because optimum temperature is between 60-75°C (1)		2

Total for this question = 9 marks

Type MS · Read an enzyme optimum from data and judge how reliable it is

Answer 4 of 11 · Question 4 part c · 2 marks

2020 NOV · Paper 02 (regional R) · extract 2 of 2 · find part c in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Read an enzyme optimum from data and judge how reliable it is

Answer 5 of 11 · Question 3 part b.iii · 3 marks

2021 JUN · Paper 01 · extract 1 of 2 · find part b.iii in this mark scheme extract

number				
3 (a)	(i)	A (amino acids); B and D are components of lipids C is a carbohydrate		1
	(ii)	• add Biuret solution to test solution; • mauve/lilac, protein present; • remains blue, protein absent;		3
(b)	(i)	• suitable axes labels; • suitable scale; • correct plots; • suitable curves; • curves labelled;	max 3 for bar chart accept key	5
	(ii)	• denaturation; • loss of shape of active site/enzyme; • substrate no longer fits/decreases rate of activity;		3
	(iii)	any three from • B is much more active/works faster; • over whole temperature range; • higher optimum temperature for B/works fastest at 80°C; • active over a greater range of temperatures;		3

Total 15 marks

Type MS · Read an enzyme optimum from data and judge how reliable it is

Answer 5 of 11 · Question 3 part b.iii · 3 marks

2021 JUN · Paper 01 · extract 2 of 2 · find part b.iii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Read an enzyme optimum from data and judge how reliable it is

Answer 6 of 11 · Question 1 part b.i · 1 mark

2021 NOV · Paper 01 · extract 1 of 2 · find part b.i in this mark scheme extract

Question number				
1	(a)	(i)	<table><tr><th>Conditions</th><th>Result/no/scale of bubbles</th></tr><tr><td>Room temperature</td><td>5</td></tr><tr><td>Alkali added</td><td>3</td></tr><tr><td>Acid added</td><td>2</td></tr><tr><td>3°C</td><td>1</td></tr><tr><td>70°C</td><td>0</td></tr><tr><td>no hydrogen peroxide added</td><td>0</td></tr></table> • two column table; • headings; • correct order of results (high to low/ORA);	3 marks
		(ii)	• no repeats/only done once;	1 mark
		(iii)	• results based on judgement/subjective/qualitative;	1 mark
		(iv)	• repeats; • measure volume of oxygen; • using gas syringe/other suitable apparatus;	3 marks
	(b)	(i)	untreated/room temperature;	1 mark
		(ii)	control;	1 mark
		(iii)	• reduced/low enzyme activity; • not at optimum pH; • active site changed; • fewer enzyme substrate complexes/substrate won't fit;	4 marks

Total 14 marks

Type MS · Read an enzyme optimum from data and judge how reliable it is

Answer 6 of 11 · Question 1 part b.i · 1 mark

2021 NOV · Paper 01 · extract 2 of 2 · find part b.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Read an enzyme optimum from data and judge how reliable it is

Answer 7 of 11 · Question 7 part a.ii · 1 mark

2023 NOV · Paper 02 · extract 1 of 2 · find part a.ii in this mark scheme extract

7 (a)	(i)	• volume/concentration of enzyme/lipase; • volume/concentration of indicator; • volume/same type of olive oil; • volume/mass of bile saltes	Ignore amount for all mps	3
	(ii)	40°C;		1
	(iii)	no values given between 30°C and 40°C; no values given between 40°C and 50°C; optimum could be between temperature intervals given/below/above 40ºC/50ºC;		3
(b)	(i)	• tube 3 = 11.5 hours and tube 8 = 24 hours; • 24 - 11.5 = 12.5 hours;	Ecf incorrect values in a subtraction calculation with a correct answer 1 mark	2
	(ii)	• lipase digests/breaks down olive oil; • produces fatty acids; • these acids lower the pH;		3
(c)		• no bile salts so no emulsification of oil/oil not broken down to droplets; • less surface area (for lipase to work on); • lipase only working very slowly/not very active/takes longer to digest oil;		3

Total 15 marks

Type MS · Read an enzyme optimum from data and judge how reliable it is

Answer 7 of 11 · Question 7 part a.ii · 1 mark

2023 NOV · Paper 02 · extract 2 of 2 · find part a.ii in this mark scheme extract

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Type MS · Read an enzyme optimum from data and judge how reliable it is

Answer 8 of 11 · Question 7 part e · 2 marks

2024 NOV · Paper 02 · extract 1 of 2 · find part e in this mark scheme extract

7 (a)	• body temperature; • optimum/enzyme is not denatured;	Accept alimentary for MP1	2
(b)	• effect of pH/to see if pH affects enzyme activity; • range of pH used/different pH used/ pH is the independent variable;	***Effect of different pH on enzyme activity = 2 marks***	2
(c)	Any four from: Answers about validity(MP1-4) • The same temperature; • The same volume of enzyme/suspension; • not (valid) because different time used; • the same enzyme is used; Answers about accuracy(MP 5-7) • as difficult to judge endpoints/ when it goes clear; • students' judgement of endpoint may vary/ subjective/ not standardised/ did not use a colorimeter; • timing is too short to reach endpoint/ cannot tell it will go clear if left for longer;	They do not have to mention validity/accuracy to score marks. Must have at least one mark from each section to score 4. Maximum of 3 if answers are from one section	4
(d)	• tube with no enzyme/ denatured enzyme/boiled enzyme; • same volume of boiled enzyme/water /5cm^3 of water/boiled enzyme; • same volume of suspension/same overall volume/ same volume of substrate (in test tube);		3
(e)	• optimum pH; • fastest between 2 and 4;		2
(f)	• stomach; • acidic/low pH/pH2-4/pH2;		2

Total 15 marks

Type MS · Read an enzyme optimum from data and judge how reliable it is

Answer 8 of 11 · Question 7 part e · 2 marks

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Type MS · Read an enzyme optimum from data and judge how reliable it is

Answer 9 of 11 · Question 7 part d · 5 marks

2019 JUN · Paper 01 (regional R) · extract 1 of 3 · find part d in this mark scheme extract

7(a)	Any four from: • volume of lactose/sugar solution used; • concentration of lactose/sugar solution; • volume of enzyme solution used; • concentration of enzyme; • type of enzyme used; • how to measure amount/concentration of lactic acid / apparatus to measure amount / concentration of lactic acid;	Allow amount	**(4)**

Question Number	Answer	Additional guidance	Mark
7(b)(i)	25 °C;	accept 20 °C to 45 °C	**(1)**

Question Number	Answer	Additional guidance	Mark
7(b)(ii)	Any two from: • too few results/temperatures investigated / temperatures are too far apart/large gaps between temperatures; • peak/maximum lactic acid production could be below 25°C / could be above 25°C;		**(2)**

Question Number	Answer	Additional guidance	Mark
7(c)	Any three from: • little/no increase in lactic acid concentration / lactic acid concentration would reach a maximum/peak/become constant; • enzyme denatures (due to decrease in pH); • all substrate converted/broken down/used up; • so reaction /slows down/stops;		**(3)**

Type MS · Read an enzyme optimum from data and judge how reliable it is

Answer 9 of 11 · Question 7 part d · 5 marks

2019 JUN · Paper 01 (regional R) · extract 2 of 3 · find part d in this mark scheme extract

Question Number	Answer	Additional guidance	Mark
7(d)	Any five from: • not possible (to be completely certain); • due to (low) number of pH/temperature values tested/more pH/temperature values need to be tested; • pH5 appears to be optimum pH; • because most sugar broken down/converted at this pH; • optimum pH could be between 5-8/2-5; • optimum temperature not known; • no repeats carried out/no means calculated; • anomalies not identified; • there could be other components in milk that increase enzyme activity/control variables not taken into account;	Allow value stated within ranges given	**(5)**

Total 15 Marks

Type MS · Read an enzyme optimum from data and judge how reliable it is

Answer 9 of 11 · Question 7 part d · 5 marks

2019 JUN · Paper 01 (regional R) · extract 3 of 3 · find part d in this mark scheme extract

Paper 1: Subject Human Biology

Question Number	Answer	Additional guidance	Mark

Type MS · Read an enzyme optimum from data and judge how reliable it is

Answer 10 of 11 · Question 7 part a.iii · 3 marks

2023 NOV · Paper 02 · extract 1 of 2 · find part a.iii in this mark scheme extract

7 (a)	(i)	• volume/concentration of enzyme/lipase; • volume/concentration of indicator; • volume/same type of olive oil; • volume/mass of bile saltes	Ignore amount for all mps	3
	(ii)	40°C;		1
	(iii)	no values given between 30°C and 40°C; no values given between 40°C and 50°C; optimum could be between temperature intervals given/below/above 40ºC/50ºC;		3
(b)	(i)	• tube 3 = 11.5 hours and tube 8 = 24 hours; • 24 - 11.5 = 12.5 hours;	Ecf incorrect values in a subtraction calculation with a correct answer 1 mark	2
	(ii)	• lipase digests/breaks down olive oil; • produces fatty acids; • these acids lower the pH;		3
(c)		• no bile salts so no emulsification of oil/oil not broken down to droplets; • less surface area (for lipase to work on); • lipase only working very slowly/not very active/takes longer to digest oil;		3

Total 15 marks

Type MS · Read an enzyme optimum from data and judge how reliable it is

Answer 10 of 11 · Question 7 part a.iii · 3 marks

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Type MS · Read an enzyme optimum from data and judge how reliable it is

Answer 11 of 11 · Question 7 part c · 4 marks

2024 NOV · Paper 02 · extract 1 of 2 · find part c in this mark scheme extract

7 (a)	• body temperature; • optimum/enzyme is not denatured;	Accept alimentary for MP1	2
(b)	• effect of pH/to see if pH affects enzyme activity; • range of pH used/different pH used/ pH is the independent variable;	***Effect of different pH on enzyme activity = 2 marks***	2
(c)	Any four from: Answers about validity(MP1-4) • The same temperature; • The same volume of enzyme/suspension; • not (valid) because different time used; • the same enzyme is used; Answers about accuracy(MP 5-7) • as difficult to judge endpoints/ when it goes clear; • students' judgement of endpoint may vary/ subjective/ not standardised/ did not use a colorimeter; • timing is too short to reach endpoint/ cannot tell it will go clear if left for longer;	They do not have to mention validity/accuracy to score marks. Must have at least one mark from each section to score 4. Maximum of 3 if answers are from one section	4
(d)	• tube with no enzyme/ denatured enzyme/boiled enzyme; • same volume of boiled enzyme/water /5cm^3 of water/boiled enzyme; • same volume of suspension/same overall volume/ same volume of substrate (in test tube);		3
(e)	• optimum pH; • fastest between 2 and 4;		2
(f)	• stomach; • acidic/low pH/pH2-4/pH2;		2

Total 15 marks

Type MS · Read an enzyme optimum from data and judge how reliable it is

Answer 11 of 11 · Question 7 part c · 4 marks

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